

Day 6: Introduction to Network Devices

Task 1 — Network Devices and Their Functions

Device	Main Function	Typical Ports / Connections
Switch	Connects multiple devices in a local network and forwards data to the correct device.	Ethernet ports, Console port, Power
Router	Connects different networks and sends data between the local network and Internet.	WAN Ethernet, LAN Ethernet, Console, Power
Hub	Connects multiple devices and sends incoming data to all connected ports.	Ethernet ports, Power
Firewall	Monitors and controls network traffic based on security rules.	WAN/LAN Ethernet ports, Console, Power
Access Point	Provides wireless Wi-Fi connectivity to devices.	Ethernet port, Power, Antenna/Wireless

Short Explanation

In a home Wi-Fi setup, the **router** normally connects the home network to the Internet. An **access point** provides Wi-Fi connectivity, while a **switch** can be used to connect multiple wired devices. A firewall helps control and protect network traffic. A hub is an older and simpler device that broadcasts data to all its ports.

Task 2 — Identify Devices and Ports

For this task, I identified three common network devices and their important ports.

1. Router

Ports to label:

- WAN/Internet Ethernet port
 - LAN Ethernet ports
 - Power port
- Console port (if available)

2. Switch

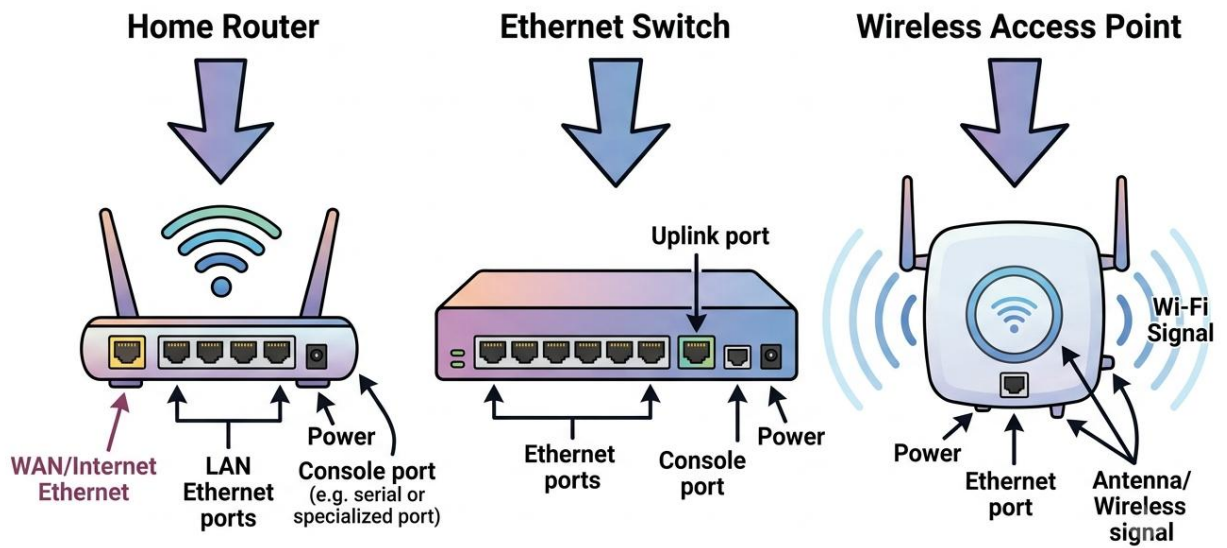
Ports to label:

- Ethernet ports
- Console port
- Uplink port (if available)
 - Power connection

3. Wireless Access Point

Ports to label:

- Ethernet port
 - Power port
- Antenna / internal wireless antenna



Result:

I learned how to identify common network devices and their main physical ports.

Task 3 — Network Devices Used with Instagram

I use **Instagram** on my phone through a Wi-Fi network.

First, my phone connects to a **wireless access point** using Wi-Fi. The access point connects the wireless device to the local network. A **router** then forwards the traffic from my home network towards the Internet and helps the data reach Instagram's servers. A firewall can also inspect and control network traffic before it reaches the Internet.

Simple Flow

**Phone → Access Point → Router/Firewall →
Internet → Instagram Server**

Task 4 — Slow College Wi-Fi

One possible reason for slow college Wi-Fi is that an **access point may have too many connected devices**. If one access point is handling too many users at the same time, the available wireless capacity is shared between many devices.

Another possible issue can be network traffic or an overloaded router/switch.

Possible Solution

Upgrading or adding **higher-capacity access points** and distributing users across multiple access points could improve Wi-Fi coverage and reduce the number of devices handled by each access point.

For example, instead of one access point serving a large classroom and corridor, multiple properly placed access points can provide better coverage and handle more users.

Result:

The problem can be related to device capacity,

coverage or traffic. Improving the access point capacity and placement can help provide a more stable Wi-Fi connection.

Conclusion

In this assignment, I learned about switches, routers, hubs, firewalls and wireless access points. I also learned about their functions and common ports. Network devices work together to connect users, control traffic and provide access to the Internet.